

2022 JPJC MYE

Paper 1 (Written)

(a)

Conditions	1	2	3	4	5	6	7	8
Order amount >\$150	Y	Y	Y	Y	N	N	N	N
"Paylor!" as payment mode	Y	Y	N	N	Y	Y	N	N
Delivery address > 10km	Y	N	Y	N	Y	N	Y	N
Actions (Total shipping cost = Standard shipping + distance-based surcharge)								
Total shipping fee = \$0 + \$0 = \$0		X						
Total shipping fee = \$0 + \$8 = \$8	X							
Total shipping fee = \$10 + \$0 = \$10				X		X		X
Total shipping fee = \$10 + \$8 = \$18			X		X		X	

(b)

Conditions	1	2	3	5	7	6	8	4
Order amount >\$150	Y	Y	Y	N		N		Y
"Paylor!" as payment mode	Y	Y	N	–		–		N
Delivery address > 10km	Y	N	Y	Y		N		N
Actions (Total shipping cost = Standard shipping + distance-based surcharge)								
Total shipping fee = \$0 + \$0 = \$0		X						
Total shipping fee = \$0 + \$8 = \$8	X							
Total shipping fee = \$10 + \$0 = \$10						X		X
Total shipping fee = \$10 + \$8 = \$18			X	X				

(a) (i) $2^7 = 128$ unique representations

(ii) **[1m for one point]:**

- Limited representations as it can only represent 128 distinct characters. Therefore can only represent English characters and not other languages.
- Binary arithmetic for the denary digits characters cannot be applied directly as the number digit character is not represented by its actual binary value.

(iii)

$$\begin{aligned}
 \text{ASCII}('A') &= 65_{10} = 41_{16} & \text{ASCII}('Z') &= 90_{10} = 5A_{16} \\
 41_{16} &= 4_{16} \quad 1_{16} & 5A_{16} &= 5_{16} \quad A_{16} \\
 &= 0100 \ 0001_2 & &= 0101 \ 1010_2 (8\text{-bit})
 \end{aligned}$$

(iv)

Unicode can support a greater range of characters representation than ASCII.

It can represent the characters of more than one language, mathematical symbols and even emojis

- (b)** Group the 48 bits into groups of 8 bits starting from LSB
- Group the 48 bits into groups of 8 bits starting from LSB
 - In each group of 8 bits, group bits into 2 groups of 4 bits, and
 - then convert the 4 bits into hexadecimal

001110000110100010010011101110111001110000000001₂

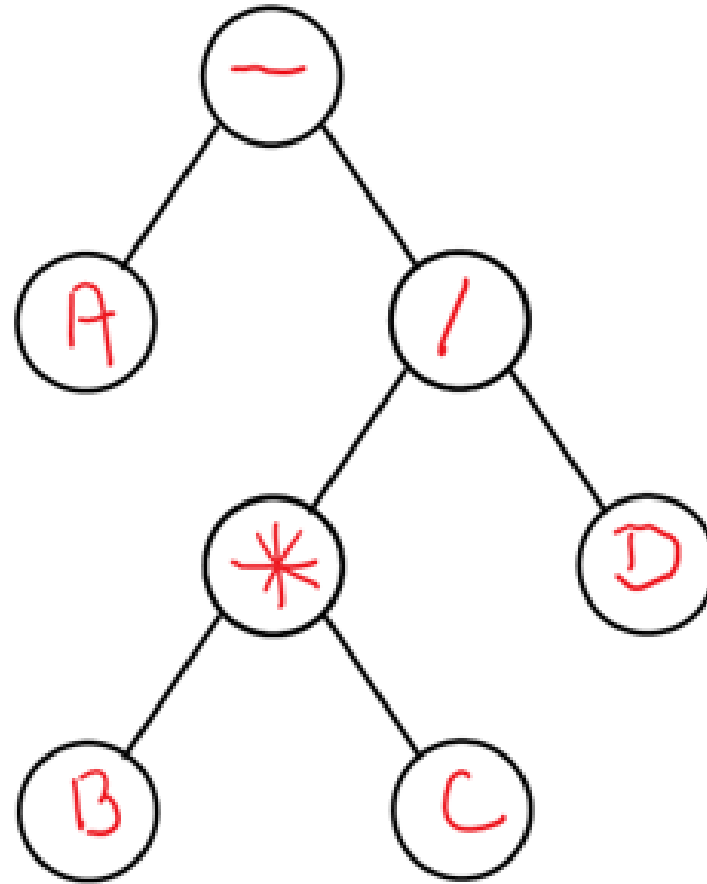
(0011 1000) : (0110 1000) : (1001 0011) : (1011 1011) : (1001 1100) : (0000 0001₂)

38 : 68 : 93 : BB : 9C : 01₁₆

(a) A binary tree:

- Is abstract data type (ADT) that consists a **finite set of nodes** as a **non-linear/hierachical** data structure,
- Every **node has at most 2 subtrees/child nodes** (one or both can be non-existent)
- Each a child node is labelled as being either a 'left-child', or a 'right-child', where a **left child precedes its sibling right child** in the order of children of a node

(b)



(c) A recursive procedure can:

- Call itself from the main body over and over, **[1m]**
- Can terminate and return control to its caller when the base case is reached. **[1m]**

(d) Recursion is when a subroutine **calls itself within its code**, thus **repeatedly executing the sequence of instructions inside it**.

An iteration subroutine **uses a loop to execute the set of instructions repeatedly** using **"FOR", "WHILE", and "REPEAT UNTIL"** loops.

(e) A run-time stack/program stack/stack **[1m]**

- Whenever procedure P is called (whether from the main program or when it calls itself), a new stack frame that contains all the procedure's data will be created and pushed into the top of the program stack, with the stack pointer pointing to it. The data stored in the stack frame includes its:
 - return address
 - value(s) of its input parameter(s),
 - value(s) of the local variable(s),
 - saved values of registers.
- When the current procedure P has completed its tasks, control will need to be passed back to its caller function. **[1m]**
- This can be done by removing the current stack frame where the stack pointer is pointing by performing a pop operation on the program stack, thus moving the stack pointer to the stack frame below, which is the program stack of its caller function. **[1m]**

(f) (i) P calls itself 6 times. **[1m]**

P(leftPtr[3] = 4)	P(rightPtr[6] = 2)
Prints('A')	Print('C')
P(rightPtr[3] = 5)	Print('*')
P(leftPtr[5] = 6)	P(rightPtr[5] = 2)
P(leftPtr[6] = 1)	Print('D')
Print('B')	Print('/')

(ii) Final output: "ABC*D/" **[1m]**

(g) Procedure P prints out the data stored in the node in postorder traversal.

- (a) A stack is a **First In Last Out data structure**, where the only item that can be access is the top item the stack pointer is pointing to. **[1m]**

Adding a new item into the stack requires it to be **pushed into the top** of the stack with the **stack pointer pointing** to it. **[1m]**

Deletion of an item can only be performed on the existing topmost item in the stack referenced by the stack pointer. **[1m]**

- (b) Advantages:

- Easy to implement
- Supports backtracking algorithm like recursive functions

Disadvantages:

- Unable to scale (cannot increase/decrease size) which may lead to inefficient utilisation of memory.
- Unable to access all items in the stack in one traversal (no searching).

(c) Stacks can be used for

- inorder, preorder, postorder traversals of a binary tree.
- Evaluating postfix algebraic expressions,
- Memory management
- Implementing backtracking algorithms (recursive algorithms).

(d)

```
PROCEDURE Push(item):
```

```
    IF topPointer = 10 THEN [1m: check full]
```

```
        PRINT('Stack is full. Item not added')
```

```
    ENDIF
```

```
    topPointer  $\leftarrow$  topPointer + 1 [1m: update pointer and store]
```

```
    ar[topPointer]  $\leftarrow$  item
```

```
ENDPROCEDURE
```

```
FUNCTION POP():
```

```
    IF topPointer = -1 THEN [1m: check empty]
```

```
        PRINT("Stack is empty. No item to return")
```

```
    ENDIF
```

```
    DECLARE temp : INTEGER
```

```
    temp  $\leftarrow$  topPointer [1m: update pointer and return ar[temp]
```

```
    topPointer  $\leftarrow$  topPointer - 1
```

```
    RETURN ar[temp]
```

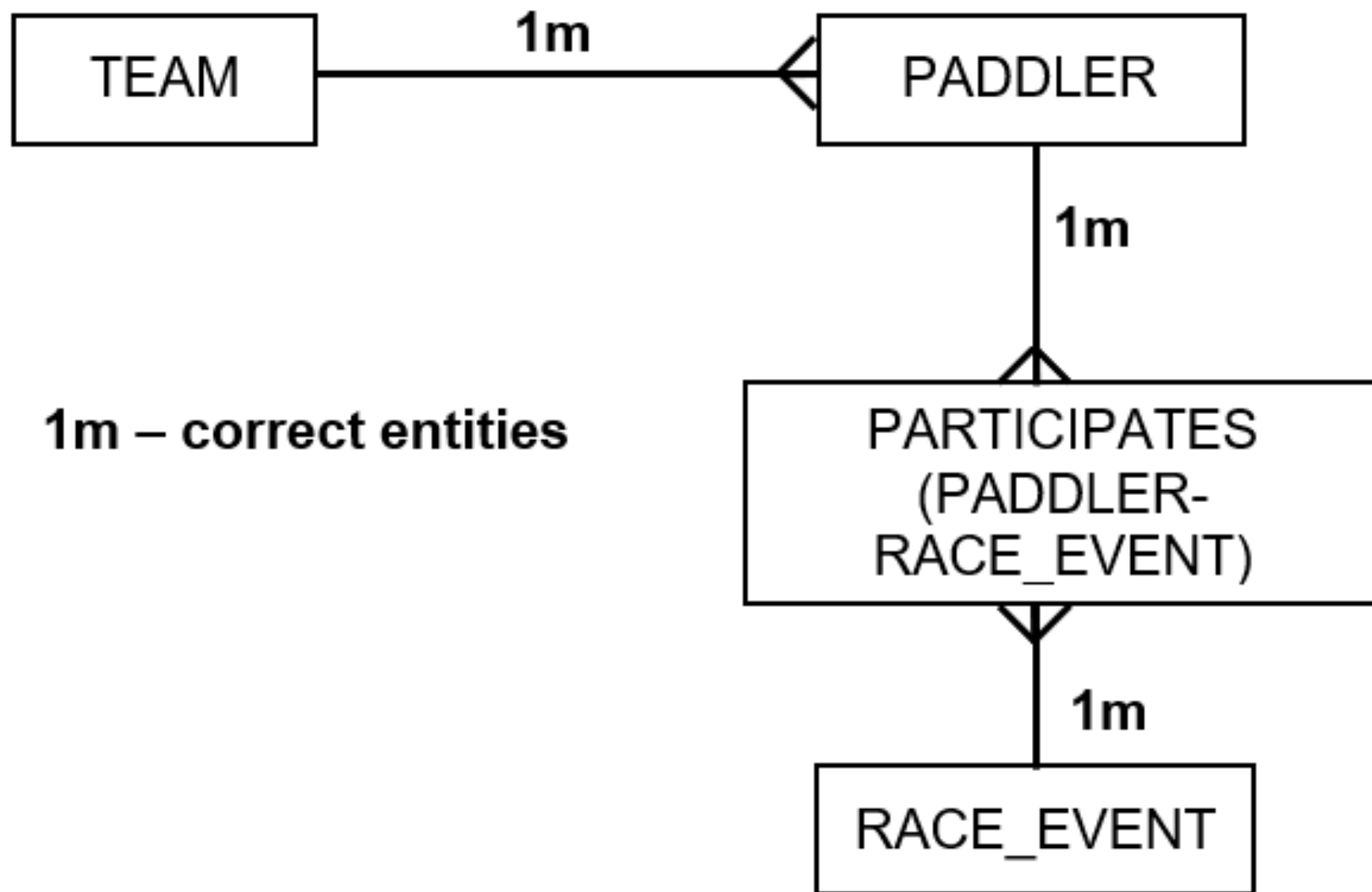
```
ENDFUNCTION
```

- (a) A key-value pair consists of **2 data elements that are taken from the same data item** where the key is used as a unique identifier to the remaining values of the data.
- (b) - A **hash algorithm** is applied to the **key of a key-value pair**. [1m]
- The hash algorithm will perform some arithmetic transformation to the key in order to **obtain a hash address**. [1m]
 - The values associated to the key will be **stored in an array at the location determined by the hash address**. [1m]

- (c) A collision is an event when the **hash address of a key-value pair has already been occupied by another key-value pair** that was added into the hash table before it.
- (d) Benefit: Accessing the key-value pairs in hash table is its **speed/fast time taken**. (takes $O(1)$ lookup time) **[1m]**

Reason: **Array allows direct access** to its data element. **[1m]**

(a)



TEAM(TeamName, ManagerName, ManagerContact,
ManagerEmail)

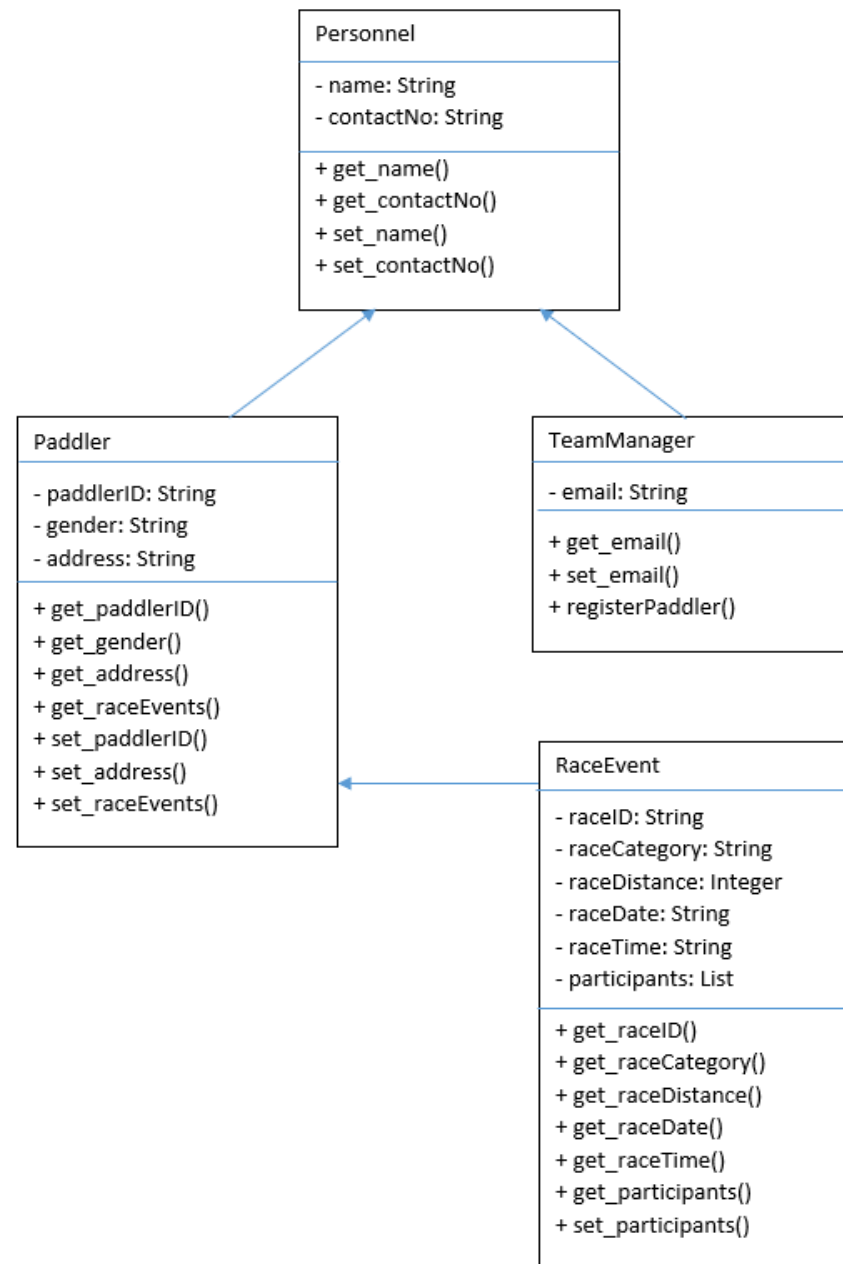
PADDLER(paddlerID, name, gender, contactNo, address)

RACE_EVENT(raceID, raceCategory, raceDistance,
raceDate, raceTime)

PARTICIPATES(paddlerID*, raceID*)


```
SELECT RACE_EVENT.raceID, RACE_EVENT.raceCategory,  
RACE_EVENT.raceDate,  
RACE_EVENT.raceTime, RACE_EVENT.raceDistance  
  
FROM RACE_EVENT, PARTICIPATES  
WHERE PARTICIPATES.paddlerID = paddlerID AND  
PARTICIPATES.raceID = RACE_EVENT.raceID
```

Q6



The purpose of a superclass is to allow **reusable code** by allowing subclasses to **inherit its attributes and methods** while extending them further.



(a) Binary search

(b) The elements in array `ar` must be **pre-sorted into some order**.

(c)

REPEAT UNTIL loop will first perform its task **at least once** since the condition(s) to break out of the loop is **inspected after the end of its repetition**

WHILE loop will first **inspect condition(s) before it executes its repetition**.

Hence the number of times it **repeats itself can be 0 or more times**.

(d) `name > ar[mid]`

(e)

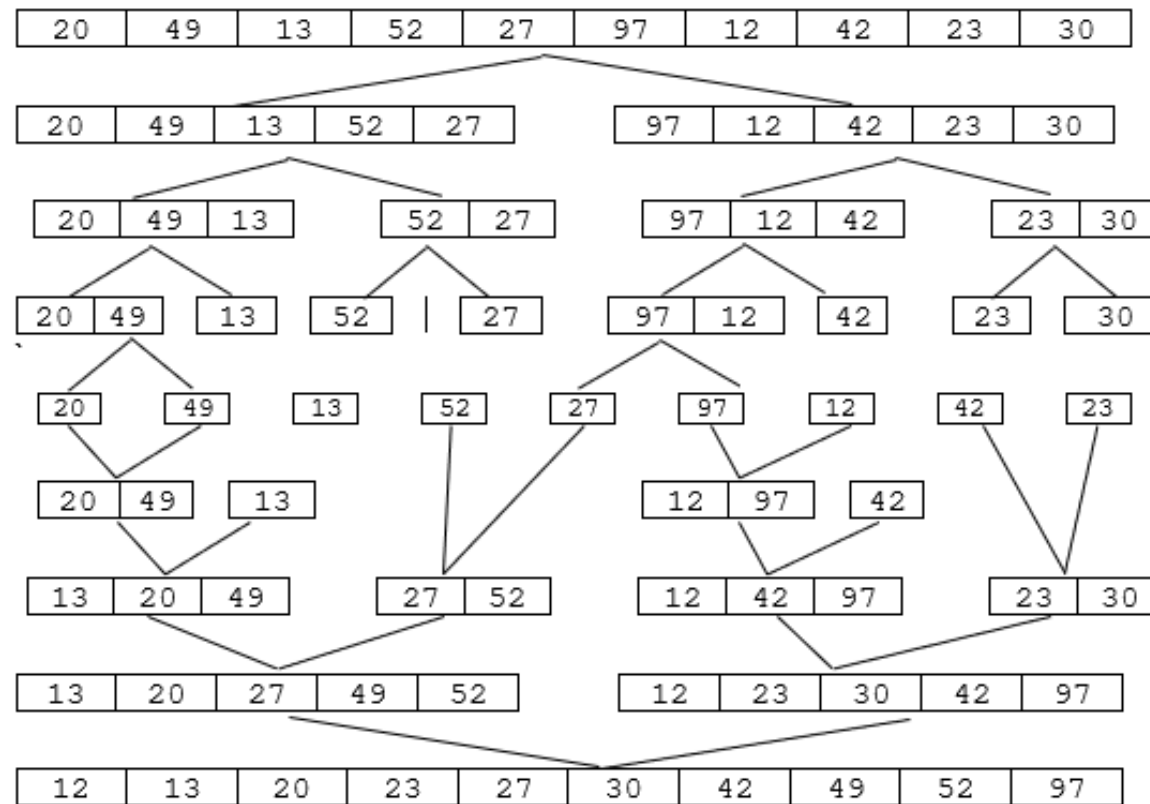
using meaningful variable, procedure, and function names.

Proper indentation

Use of comments.

(f)

first	mid	last		
1	84	168	name < ar[mid]	Last = 84-1
1	42	83	name < ar[mid]	Last = 42-1
1	21	41	Name > ar[mid]	First = 21+1
22	31	41	Name < ar[mid]	Last = 31-1
22	26	30	Name = ar[mid]	FOUND



1. By finding the middle index of the unsorted array and **recursively divide the unsorted array into 2 halves** until multiple subarrays of single element remains.
2. Compare by sorting and merging the single element subarrays from bottoms up into pairs.
3. Recursively sort and merge the elements of the subarrays to produce new sorted subarrays until a single sorted array is formed eventually.
4. **Recursively sort and merge starting with the single element subarrays to produce sorted subarrays.**
5. **Terminate recursion when a single array of multiple elements is formed.**

- (h)
- (i) Time complexity for merge sort = $O(n \log_2 n)$ **[1m]**
 - (ii) The time complexity of MergeSort is $O(n \log_2 n)$ in all the 3 cases (worst, average and best) as the mergesort will **always divide the dataset into two halves [1m]**, and **takes linear time to merge two halves regardless of how sorted the elements are in the dataset. [1m]**

- (a) A wide area network (WAN) is a **large computer network** that connects groups of computers **over large distances**. [1m]

WANs are often used by large businesses to connect their office networks; each office typically has its own local area network, or LAN, and these LANs connect via a WAN.

- (b) A protocol is a **set of rules and standards that govern the communications between 2 or more devices connected in a network.**

Protocols are important in data communication because in order for both the sender and receiver to understand and successfully send/receive all the data packets, they must **first agree upon a set of rules to be used throughout their communication**

- (c) When a host system initially sends a packet, it **looks up the packet's destination address in its routing table to determine if the destination is on the local network**. If yes, the packet goes directly to the host with that IP address. If not, the packet goes to a router on the local network.

When an intermediate router receives a packet, the router checks its routing table to determine **if the destination address is for a system on one of its attached networks or if the message must be forwarded through another router**. It then sends the message to the next system in the path to the destination

This **process is repeated on each router that receives the message until the message reaches the destination system**

(a) Any one of the possible reasons:

Happy Supermarket has sold the data collated taken from the filled-up lucky draw tickets to other companies as sales contacts.

Data collated taken from the filled-up lucky draw tickets could have been misplaced/stolen by unauthorised personnel.

The system that records the data collated taken from the filled-up lucky draw tickets could have been hacked.

Or any other reasonable points written in the context of the question.

(b)

- **Consent:** Happy Supermarket must obtain Mdm Liu's knowledge and consent to collect, use or disclose his/her personal data (with some exceptions).
- **Notification:** Happy Supermarket must inform individuals of the purposes for collecting, using or disclosing their personal data.
- **Appropriateness:** Happy Supermarket **may only collect, use or disclose** personal data only for **purposes that would be considered appropriate to a reasonable person** under the given circumstances.
- **Accountability:** Happy Supermarket must make information about their personal data protection policies available on request. The contact details of Happy Supermarket representatives responsible for answering questions relating to the organisations' collection, use or disclosure of personal data should also be made known to the public.